Variable Selection for Wasserstein Robust Hypothesis Testing

Abstract

We derive an exact finite convex formulation for Wasserstein robust testing with coordinate selection. A uniform risk bound introduces the selection variables without changing binary optimal values. The resulting convex value function admits affine dual cuts for optimization over coordinate subsets.

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we follow the convention in [2] to define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \leq \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \leq d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Finite reformulation

Assume $n_0, n_1 \geq 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. All distributions have finite first moments. Throughout, $i \in [n_0]$ indexes class 0, $j \in [n_1]$ indexes class 1, and $m \in [D]$ indexes a coordinate. For a fixed selection, define

$$V(S) = \min_{f \in \mathcal{F}_S} R(f), \qquad c_{ij}(S) = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1.$$

The least-favorable-distribution viewpoint of [2] suggests transporting empirical mass between observations. We derive the finite problem directly for the generating loss in Section 1. For observation $r \in [n_k]$ of class k, introduce an auxiliary score $u_r^{(k)}$ and a source loss bound $a_r^{(k)}$; let $\lambda_k \geq 0$ price the transport budget of class k. Write

$$J(a, \lambda) = \frac{1}{n_0} \sum_i a_i^{(0)} + \frac{1}{n_1} \sum_j a_j^{(1)} + \rho_0 \lambda_0 + \rho_1 \lambda_1.$$

Proposition 2.1 (Finite reformulation)**.** *For every $S \subseteq [D]$, the following convex problem attains $V(S)$:*

$$\begin{aligned}
\min_{a,u,\lambda} \quad & J(a,\lambda) \\
\text{subject to} \quad & \ell(-u_i^{(0)}) \le a_i^{(0)}, && i \in [n_0], \\
& \ell(u_j^{(1)}) \le a_j^{(1)}, && j \in [n_1], \\
& \ell(-u_j^{(1)}) \le a_i^{(0)} + \lambda_0 c_{ij}(S), && i \in [n_0],\ j \in [n_1], \\
& \ell(u_i^{(0)}) \le a_j^{(1)} + \lambda_1 c_{ij}(S), && i \in [n_0],\ j \in [n_1], \\
& a^{(k)}, u^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \ge 0, && k = 0, 1.
\end{aligned} \tag{3}$$

An optimal solution determines a measurable detector on the whole observation space with risk $V(S)$.

The first two constraints bound the losses of mass that stays at its source. The last two bound transfers to the opposite class. For example, class-0 mass moved from $x_i^{(0)}[S]$ to $x_j^{(1)}[S]$ has loss $\ell(-u_j^{(1)})$, bounded by $a_i^{(0)}$ plus the priced cost $\lambda_0 c_{ij}(S)$. Appendix A proves exactness in both directions: these restricted transports give a lower bound on the original risk, and every feasible finite solution gives a detector whose risk is at most its objective. This also covers coincident projections.

The zero detector has risk 2. Since the diagonal constraints imply $a^{(k)} \ge 0$, an optimum satisfies

$$\rho_0\lambda_0 + \rho_1\lambda_1 \le J(a,\lambda) = V(S) \le 2. \tag{4}$$

This bound will let a binary variable switch a coordinate on or off.

3 Selection variables and duality

Let $z = \mathbf{1}_S$, and define

$$\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}, \qquad \Delta_{ij,m} = |x_i^{(0)}[m] - x_j^{(1)}[m]|.$$

Then $c_{ij}(S) = \sum_m \Delta_{ij,m} z_m$. Introduce w_{km} to replace the products $\lambda_k z_m$. The constraints $0 \le w_{km} \le \lambda_k$ and $\rho_0 w_{0m} + \rho_1 w_{1m} \le 2z_m$ force $w_{0m} = w_{1m} = 0$ when $z_m = 0$. When $z_m = 1$, the choice $w_{km} = \lambda_k$ is permitted at an optimum by (4). For $z \in [0,1]^D$, define

$$\begin{aligned}
F(z) = \min_{a,u,\lambda,w} \quad & J(a,\lambda) \\
\text{subject to} \quad & \ell(-u_i^{(0)}) \le a_i^{(0)}, \quad \ell(u_j^{(1)}) \le a_j^{(1)}, \\
& \ell(-u_j^{(1)}) \le a_i^{(0)} + \sum_m \Delta_{ij,m} w_{0m}, \\
& \ell(u_i^{(0)}) \le a_j^{(1)} + \sum_m \Delta_{ij,m} w_{1m}, \\
& 0 \le w_{km} \le \lambda_k, \qquad k = 0,1,\ m \in [D], \\
& \rho_0 w_{0m} + \rho_1 w_{1m} \le 2z_m, \qquad m \in [D], \\
& a^{(k)}, u^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \ge 0, \qquad k = 0,1.
\end{aligned} \tag{5}$$

The loss constraints hold for every $i \in [n_0], j \in [n_1]$. This has the two-stage logical structure of problems (2.1)–(2.2) in [1]: each coordinate activates the pair (w_{0m}, w_{1m}) of a convex continuous problem, and the outer problem chooses at most d such pairs. The bound 2 follows from the risk; it is not a tuning parameter.

Proposition 3.1 (Selection formulation)**.** *The minimum in* (5) *is attained. The function F is convex on $[0,1]^D$, and*

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S| \le d} V(S) = \min_{z \in \mathcal{Z}_d} F(z). \tag{6}$$

Proof. An optimum of (3) extends to (5) by $w_{km} = \lambda_k z_m$ at $z = \mathbf{1}_S$, using (4). Conversely, feasibility at that binary point implies $w_{km} = 0$ outside S and

$$\sum_m \Delta_{ij,m} w_{km} \le \lambda_k \sum_{m \in S} \Delta_{ij,m} = \lambda_k c_{ij}(S).$$

It therefore implies feasibility in (3) with the same objective. This proves equality of the binary optimal values. For any z, the zero detector gives a feasible objective of 2. On $J \le 2$, the source bounds and prices are bounded; both signs of each score are bounded as in Appendix A, and $0 \le w_{km} \le \lambda_k$. Thus an optimum exists. All constraints in (5) are jointly convex, including in z. A convex combination of optimal solutions at z, z' is feasible at $tz + (1-t)z'$, so

$$F(tz + (1-t)z') \le tF(z) + (1-t)F(z'), \qquad 0 \le t \le 1.$$

□

3.1 Dual representation

Let $\Gamma^{(0)}_{ij} \ge 0$ be the multiplier of the class-0 cross constraint and $\Gamma^{(1)}_{ij} \ge 0$ that of the class-1 cross constraint. They represent mass transported from source i to target j, and from source j to target i, respectively. Write $\Gamma = (\Gamma^{(0)}, \Gamma^{(1)})$. At a location with class masses $p, q \ge 0$, the best weighted loss is

$$h(p,q) = \inf_{v \in \mathbb{R}} \{p\ell(-v) + q\ell(v)\}.$$

At a class-0 observation, the remaining class-0 mass is $1/n_0 - \sum_j \Gamma^{(0)}_{ij}$, and the incoming class-1 mass is $\sum_j \Gamma^{(1)}_{ij}$. Applying the same reasoning to class 1 gives the total optimized loss

$$\begin{aligned} H(\Gamma) = &\sum_i h\left(\frac{1}{n_0} - \sum_j \Gamma^{(0)}_{ij}, \sum_j \Gamma^{(1)}_{ij}\right) \\ &+ \sum_j h\left(\sum_i \Gamma^{(0)}_{ij}, \frac{1}{n_1} - \sum_i \Gamma^{(1)}_{ij}\right). \end{aligned} \tag{7}$$

Let $\mu_m \ge 0$ multiply the coordinate constraint $\rho_0 w_{0m} + \rho_1 w_{1m} \le 2z_m$, and set

$$\mathcal{D} = \left\{ (\Gamma, \mu) \ge 0 : \begin{array}{ll} \sum_j \Gamma^{(0)}_{ij} \le 1/n_0 & (i \in [n_0]), \\ \sum_i \Gamma^{(1)}_{ij} \le 1/n_1 & (j \in [n_1]), \\ \sum_m \left(\sum_{i,j} \Gamma^{(k)}_{ij} \Delta_{ij,m} - \rho_k \mu_m \right)_+ \le \rho_k & (k = 0, 1) \end{array} \right\}, \tag{8}$$

where $(v)_+ = \max\{v, 0\}$ and every sum uses the index ranges defined above. The first two constraints limit outgoing mass. The last two arise by minimizing the Lagrangian over λ, w, as shown in Appendix B. The set $\mathcal{D}$ is independent of z.

Proposition 3.2 (Dual representation)**.** *For every $z \in [0,1]^D$, the following maximum is attained:*

$$F(z) = \max_{(\Gamma,\mu) \in \mathcal{D}} \{H(\Gamma) - 2\mu^\top z\}. \tag{9}$$

An optimizer (Γ^t, μ^t) at z^t gives the supporting function

$$\begin{aligned} c_t(z) &= H(\Gamma^t) - 2(\mu^t)^\top z \\ &= F(z^t) - 2(\mu^t)^\top (z - z^t) \le F(z), \end{aligned} \tag{10}$$

with $c_t(z^t) = F(z^t)$.

The function h is concave in its weights, so H is concave, and $\mathcal{D}$ is convex. Thus the dual at fixed z is a convex optimization problem. Its affine dependence on z gives the cutting-plane construction underlying Theorem 2.5 and Algorithm 3.1 of [1]. The capacity constraints of this model are handled directly in Appendix B.

4 Cutting-plane algorithm

Write $\mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z)$. After evaluating $z^1, \dots, z^t$ and obtaining their dual optimizers, let $U_t = \min_{s \le t} F(z^s)$ and solve

$$\begin{aligned} (z^{t+1}, L_t) \in \underset{z \in \mathcal{Z}_d,\, \eta \ge 0}{\arg\min} \quad & \eta \\ \text{subject to} \quad & \eta \ge F(z^s) - 2(\mu^s)^\top (z - z^s), \qquad s = 1, \dots, t. \end{aligned} \tag{11}$$

Every retained cut is a global lower bound and is exact at its evaluated selection. Consequently,

$$L_t \le \mathrm{OPT} \le U_t. \tag{12}$$

Starting from any $z^1 \in \mathcal{Z}_d$, alternate the convex subproblem (5)–(9) and the binary linear master (11). Stop when $U_t - L_t \le \varepsilon$ and return the best evaluated selection. Its detector is recovered by (15), with risk U_t and optimality gap at most ε. For an inexact inner solve, use the actual feasible dual expression $H(\Gamma^t) - 2(\mu^t)^\top z$ as the cut and a primal feasible objective as the upper bound; the master must provide a global lower bound. The binary formulation is exact, although its continuous relaxation can be weak.

A Proof of the finite reformulation

Proof of Proposition 2.1. Let $W(S)$ be the infimum in (3). For $f(x) = g(x[S])$, set $u_r^{(k)} = g(x_r^{(k)}[S])$. Restrict transport to retaining mass $s_i^{(0)}, s_j^{(1)} \ge 0$ at its source or moving mass $\Gamma_{ij}^{(0)}, \Gamma_{ij}^{(1)} \ge 0$ to an opposite-class selected observation, in the directions of Section 3.1:

$$\begin{aligned} s_i^{(0)} + \sum_j \Gamma_{ij}^{(0)} = 1/n_0, \qquad s_j^{(1)} + \sum_i \Gamma_{ij}^{(1)} = 1/n_1, \\ \sum_{i,j} c_{ij}(S) \Gamma_{ij}^{(k)} \le \rho_k, \qquad k = 0, 1. \end{aligned} \tag{13}$$

The class-0 objective is

$$\sum_i s_i^{(0)} \ell(-u_i^{(0)}) + \sum_{i,j} \Gamma_{ij}^{(0)} \ell(-u_j^{(1)}).$$

With free mass multipliers $a_i^{(0)}$ and budget multiplier $\lambda_0 \ge 0$, its maximization Lagrangian is

$$\begin{aligned} \frac{1}{n_0} \sum_i a_i^{(0)} + \rho_0 \lambda_0 + \sum_i s_i^{(0)} [\ell(-u_i^{(0)}) - a_i^{(0)}] \\ + \sum_{i,j} \Gamma_{ij}^{(0)} [\ell(-u_j^{(1)}) - a_i^{(0)} - \lambda_0 c_{ij}(S)]. \end{aligned} \tag{14}$$

Its supremum is finite exactly when both bracketed coefficients are nonpositive. The class-1 calculation gives the other two constraints in (3). Both transport LPs are feasible and bounded; LP duality therefore identifies their maximum risk with the minimum of $J(a, \lambda)$ at these fixed scores. Replacing only a source's selected coordinates by the target coordinates lifts each plan to the original ambiguity set, with cost $c_{ij}(S)$ and the same loss. Minimizing also over u gives $W(S) \le R(f)$, hence $W(S) \le V(S)$.

Conversely, for feasible (a, u, λ) define

$$E_k(y) = \min_{r \in [n_k]} \{a_r^{(k)} + \lambda_k \| x_r^{(k)}[S] - y \|_1\}, \qquad k = 0, 1,$$

for $y \in \mathbb{R}^{|S|}$. Choose the smallest minimizing indices $i = i(y)$, $j = j(y)$ for $E_0(y)$, $E_1(y)$, and set

$$r_0 = \|x_i^{(0)}[S] - y\|_1, \qquad r_1 = \|x_j^{(1)}[S] - y\|_1, \qquad b = c_{ij}(S).$$

Define

$$g(y) = \begin{cases} (1-\gamma)u_i^{(0)} + \gamma u_j^{(1)}, & \gamma = \min\{1, r_0/b\}, \quad b > 0, \\ u_i^{(0)}, & b = 0. \end{cases} \tag{15}$$

For $b > 0$, the triangle inequality gives $\gamma b \le r_0$ and $(1-\gamma)b \le r_1$. Convexity and the four constraints yield

$$\begin{aligned} \ell(-g(y)) &\le (1-\gamma)\ell(-u_i^{(0)}) + \gamma\ell(-u_j^{(1)}) \\ &\le a_i^{(0)} + \lambda_0\gamma b \le E_0(y), \\ \ell(g(y)) &\le (1-\gamma)\ell(u_i^{(0)}) + \gamma\ell(u_j^{(1)}) \\ &\le a_j^{(1)} + \lambda_1(1-\gamma)b \le E_1(y). \end{aligned} \tag{16}$$

For $b = 0$, the class-0 diagonal and class-1 cross constraints give the same bounds. The finite tie rule makes g Borel measurable, and $|g| \le \max_{k,r} |u_r^{(k)}|$. For every full-space input x and source,

$$\begin{aligned} \ell(-g(x[S])) &\le a_i^{(0)} + \lambda_0\|x_i^{(0)}[S] - x[S]\|_1 \le a_i^{(0)} + \lambda_0\|x_i^{(0)} - x\|_1, \\ \ell(g(x[S])) &\le a_j^{(1)} + \lambda_1\|x_j^{(1)}[S] - x[S]\|_1 \le a_j^{(1)} + \lambda_1\|x_j^{(1)} - x\|_1. \end{aligned}$$

For $\delta > 0$ and any $\mathbb{P}_k \in \mathcal{P}_k$, integrate against empirical-to-$\mathbb{P}_k$ couplings of cost at most $\rho_k + \delta$:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_0}\ell(-g(X[S])) &\le \frac{1}{n_0}\sum_i a_i^{(0)} + \lambda_0(\rho_0 + \delta), \\ \mathbb{E}_{\mathbb{P}_1}\ell(g(X[S])) &\le \frac{1}{n_1}\sum_j a_j^{(1)} + \lambda_1(\rho_1 + \delta). \end{aligned}$$

Letting $\delta \downarrow 0$ and setting $f(x) = g(x[S])$ gives $R(f) \le J(a, \lambda)$. Thus $V(S) \le W(S)$, proving equality.

The feasible point $a^{(k)} = \mathbf{1}$, $u^{(k)} = 0$, $\lambda = 0$ has value 2. On $J \le 2$, $0 \le a_r^{(k)} \le 2n_k$ and $0 \le \lambda_k \le 2/\rho_k$. Each score's two losses are bounded by its diagonal and an opposite-class constraint. Since $\ell(t) \to \infty$ as $t \to \infty$, the scores are bounded. This sublevel is compact, so a minimizer exists; its recovered detector attains $V(S)$. □

B Proof of the dual representation

Proof of Proposition 3.2. Besides Γ, μ from Section 3.1, assign nonnegative multipliers $s_i^{(0)}, s_j^{(1)}$ to the diagonal loss constraints and β_{km} to $w_{km} \le \lambda_k$. Keeping a, u free and $\lambda, w \ge 0$, the minimization Lagrangian is

$$\begin{aligned} \mathcal{L} = J(a,\lambda) &+ \sum_i s_i^{(0)}[\ell(-u_i^{(0)}) - a_i^{(0)}] + \sum_j s_j^{(1)}[\ell(u_j^{(1)}) - a_j^{(1)}] \\ &+ \sum_{i,j} \Gamma_{ij}^{(0)} \left[\ell(-u_j^{(1)}) - a_i^{(0)} - \sum_m \Delta_{ij,m} w_{0m}\right] \\ &+ \sum_{i,j} \Gamma_{ij}^{(1)} \left[\ell(u_i^{(0)}) - a_j^{(1)} - \sum_m \Delta_{ij,m} w_{1m}\right] \\ &+ \sum_{k,m} \beta_{km}(w_{km} - \lambda_k) + \sum_m \mu_m(\rho_0 w_{0m} + \rho_1 w_{1m} - 2z_m). \end{aligned} \tag{17}$$

The infimum over a is finite exactly when

$$s_i^{(0)} + \sum_j \Gamma_{ij}^{(0)} = 1/n_0, \qquad s_j^{(1)} + \sum_i \Gamma_{ij}^{(1)} = 1/n_1.$$

Eliminating $s \geq 0$ gives the first two constraints of (8). The score-dependent terms are

$$\sum_i \left[s_i^{(0)} \ell(-u_i^{(0)}) + \left(\sum_j \Gamma_{ij}^{(1)} \right) \ell(u_i^{(0)}) \right] + \sum_j \left[\left(\sum_i \Gamma_{ij}^{(0)} \right) \ell(-u_j^{(1)}) + s_j^{(1)} \ell(u_j^{(1)}) \right],$$

whose infimum is $H(\Gamma)$ in (7). The coefficients of $\lambda, w \geq 0$ are

$$\lambda_k : \quad \rho_k - \sum_m \beta_{km}, \qquad w_{km} : \quad \beta_{km} + \rho_k \mu_m - \sum_{i,j} \Gamma_{ij}^{(k)} \Delta_{ij,m}.$$

They are nonnegative, with $\beta \geq 0$, exactly when

$$\sum_m \beta_{km} \leq \rho_k, \qquad \beta_{km} \geq \left(\sum_{i,j} \Gamma_{ij}^{(k)} \Delta_{ij,m} - \rho_k \mu_m \right)_+ .$$

Eliminating β gives the last two constraints of (8), leaving the dual objective $H(\Gamma) - 2\mu^\top z$.

For fixed $z \in [0,1]^D$, eliminate $w_{km} = 0$ wherever $z_m = 0$ and their cap and capacity constraints. The remaining system has the Slater point

$$u^{(k)} = 0, \qquad a^{(k)} = 2\mathbf{1}, \qquad w_{km} = \frac{z_m}{2\rho_k}, \qquad \lambda_k = 1 + \frac{1}{2\rho_k} :$$

all loss constraints are strict, $0 < w_{km} < \lambda_k$, and $\rho_0 w_{0m} + \rho_1 w_{1m} = z_m < 2z_m$. Strong duality and dual attainment follow. For eliminated coordinates, restore

$$\mu_m = \max_{k=0,1} \frac{1}{\rho_k} \sum_{i,j} \Gamma_{ij}^{(k)} \Delta_{ij,m}, \qquad \beta_{km} = 0 \quad (z_m = 0).$$

All coefficient conditions hold and the objective is unchanged; the full dual therefore attains $F(z)$.

Any $(\Gamma^t, \mu^t) \in \mathcal{D}$ gives the global lower bound $H(\Gamma^t) - 2(\mu^t)^\top z \leq F(z)$. Optimality at z^t gives equality there, hence

$$F(z) \geq F(z^t) - 2(\mu^t)^\top (z - z^t).$$

□

References

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